

Final Project

SDS 302 Spring 2026

Students will work in teams of **3** to analyze a research paper related to the course topics. The project consists of:

- Paper selection
- Written report
- Presentation

1. Paper Selection Requirements

Each team must choose one research paper clearly connected to at least one core topic from the course.

Requirements:

- The paper must be:
 - Published in a peer-reviewed conference or journal (e.g., NeurIPS, ICML, CVPR, JMLR),
 - OR a well-cited arXiv paper
- Must include methodological/technical content (not purely survey or blog-style)
- Must involve at least one model/algorithm covered in class:
 - Introduction. Statistical Learning Framework
 - Ordinary least squares. Ridge regression
 - Features. Hyperparameters and validation. MLE and MAP for regression
 - Bias-variance tradeoff. Sparse least squares. LASSO
 - Nonlinear least squares. Optimization, Gradient descent
 - Neural networks. Training neural networks
 - Logistic regression. Multivariate Gaussians
 - Gaussian discriminant analysis. Support vector machines
 - Nearest neighbor classification. K-means clustering
 - Decision trees. Random Forests. Boosting
 - Convolutional Neural Networks. CNN Architectures
- Should be appropriately challenging but understandable

Submission:

- Paper title + link
- 3–5 sentence justification:
 - Related course topics

- Why it is a good choice
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2. Written Report (10 points)

The report is shorter and focused on clarity.

Format:

- Length: 2–3 pages
- Clear structure (LaTeX recommended)

Required Sections:

1. Problem & Motivation (2 pts)

- What problem is addressed?
- Why does it matter?

2. Key Idea / Method (4 pts)

- Main model or approach
- Intuitive explanation (not heavy derivations)
- Connection to course concepts

3. Results (2 pts)

- What experiments were done?
- Key findings

4. Critical Insight (2 pts)

- One strength
- One limitation or possible improvement

3. Presentation (30 points)

Format:

- Duration: 10-25 minutes per group
- All members must participate

Expected Content:

- Problem overview
- Intuition behind the method
- Key technical idea (accessible explanation)
- Results
- Critical perspective

Grading Criteria:

1. Understanding & Depth (12 pts)

- Demonstrates strong grasp of the paper
- Explains ideas accurately
- Connects to course concepts

2. Clarity of Explanation (8 pts)

- Concepts explained intuitively
- Good use of examples/visuals
- Avoids reading slides

3. Technical Content (5 pts)

- Includes meaningful discussion of:
 - Model
 - Loss/optimization
 - Key mechanism

4. Organization & Delivery (3 pts)

- Logical flow
- Clear timing
- All members contribute

5. Q&A / Critical Thinking (2 pts)

- Ability to answer questions
- Shows independent thinking